Four Common Cyber Attacks

Man-in-the-Middle

This attack happens when a hacker puts themselves between two people or two systems that are trying to communicate.

- The hacker secretly listens to the data being sent.
- They can also change the data before it reaches the other side.
- It’s like someone standing between two people having a conversation and secretly changing the message.

This is a very dangerous and very common attack.

Brute Force Attack

You’ve probably seen this in movies:

- The hacker tries to guess a password by trying millions of combinations of letters, numbers, and symbols.
- They use special software that can test passwords extremely fast.

Strong Passwords Matter

Using **strong passwords** is important to protect your accounts from attacks and ensure your sensitive data remains safe.

- Use strong passwords
- Mix uppercase, lowercase, numbers, and symbols
- Make your password long

The stronger the password, the harder it is to crack.

Zero-Day Exploit

This one is very interesting.
A zero-day happens when a new weakness in software is discovered.

- Hackers try to attack that weakness immediately.
- At the same time, developers race to create a patch (a fix).

So it becomes a race against time:

- Hackers try to exploit it
- Developers try to fix it fast

This is why updates are important — updates often contain patches that close these security holes.

SQL Injection Attack

SQL is the language used to manage databases.
In an SQL injection attack:

- The hacker inserts malicious code into a website or form.
- This tricks the database into revealing sensitive information.

Usually, the exposed data includes:

- Usernames
- Passwords
- Personal details

It’s one of the most common attacks on websites.

Stay Safe

Protect yourself online by using **strong passwords**, **updating software regularly**, and being careful with **unsafe websites**.

1. What describes a Man-in-the-Middle attack?
A. A hacker quietly reads data sent between two users
B. A hacker blocks internet access on the network
C. A hacker replaces files stored on the device
D. A hacker resets all passwords on the system

2. What is the main idea of a brute force attack?
A. Testing many password options automatically
B. Sending emails pretending to be trusted sites
C. Installing malware through fake updates
D. Forcing a user to reveal login details

3. Why are Zero-Day exploits dangerous?
A. They attack weaknesses nobody has fixed yet
B. They affect systems only after long delays
C. They require approval from the user first
D. They stop working once the device restarts

4. What does an SQL Injection allow?
A. Forcing a database to reveal hidden data
B. Turning off the firewall on the network
C. Overloading the system with heavy traffic
D. Encrypting files and demanding payment

5. Which option helps reduce brute force risk?
A. Using long passwords with mixed characters
B. Saving all passwords in plain text files
C. Reusing one password for every account
D. Avoiding updates for login security

6. How can users reduce Zero-Day attacks?
A. Installing software updates when released
B. Using unsecured networks for browsing
C. Turning off all security protections
D. Ignoring warnings from the system